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Key Concepts: Adrenal Physiology

1. The adrenal gland consists of 3 cortical layers and a medulla. The zona glomerulosa is the outermost cortical layer and produces aldosterone. The zona fasciculata is the middle cortical layer and produces cortisol. The zona reticularis is the innermost cortical layer and produces adrenal androgens (DHEA).
2. Aldosterone, cortisol and DHEA are manufactured from the common steroid precursors of cholesterol and pregnenolone.
3. DHEA is produced by the $\Delta 5$ pathway, whereas aldosterone and cortisol are manufactured using the $\Delta 4$ pathway that requires the enzyme 3β-hydroxysteroid dehydrogenase.
4. Each biosynthetic enzyme has a synonym and gene product associated with it – enzymes can be referred to by any of these names. The enzyme name is most critical to know.
5. Aldosterone is produced in the zona glomerulosa. The enzyme 17α-hydroxylase is absent in this cortical layer, so progesterone is converted into 11-DOC by 21α-hydroxylase.
6. The enzyme aldosterone synthase catalyzes the conversion of 11-DOC via 2 separate intermediate mineralocorticoid products into aldosterone.
7. Aldosterone binds to the mineralocorticoid receptor inside the cells of the cortical collecting duct, where it stimulates the transcription of ENaC channels and Na+/K+ ATPases as well as the activity of ROMK channels. This increases Na+ reabsorption and K+ secretion.
8. Ang II stimulation of aldosterone secretion from the zona glomerulosa involves the IP3 pathway and elevation of intracellular Ca. Calcium stimulates the secretion of aldosterone and increases the activity of side chain cleavage enzyme, StAR, and aldosterone synthase. Increases in extracellular K+ stimulate aldosterone secretion by increasing intracellular Ca via depolarization of the membrane that opens voltage-gated Ca channels.
9. The zona fasciculata produces the glucocorticoids cortisol and corticosterone.
10. The biosynthetic pathway for cortisol uses the enzymes 3β-HSD, 17α-hydroxylase, 21α-hydroxylase and 11β-hydroxylase. An alternative minor pathway that bypasses 17α-hydroxylase generates 11-DOC, which has mineralocorticoid activity.
11. Cortisol secretion exhibits diurnal variation due to a circadian rhythm, with highest amounts of cortisol secretion in the morning. Stress (neurogenic and physiological) also stimulates cortisol production.
12. The hypothalamus releases CRH, which acts on the corticotropes of the anterior pituitary to release ACTH (synthesized from POMC). ACTH stimulates the zona fasciculata to secrete cortisol. Cortisol reduces the secretion of CRH and ACTH by negative feedback.
13. ACTH acutely increases the shuttling of cholesterol precursors through the biosynthetic pathway in the zona fasciculata. Chronically it increases the expression of biosynthetic enzymes and also has a trophic effect on the zona fasciculata and zona reticularis.
14. Cortisol binds to the glucocorticoid receptor, which stimulates gene transcription inside cells.
15. Cortisol has mineralocorticoid activity, but conversion to the inactive cortisone by 11β-HSD2 prevents cortisol binding to the mineralocorticoid receptor. The enzyme 11β-HSD1 converts cortisone back to cortisol.

16. Cortisol increases blood glucose by stimulating gluconeogenesis and by reducing glucose uptake into muscle and adipose tissue. It has many systemic effects, including anti-inflammatory and immunosuppressive effects.
17. The zona reticularis produces dehydroepiandrosterone, which is sulfated to prolong half-life, and small quantities of androstenedione. These are converted in peripheral tissues to testosterone and estradiol.
18. The synthesis of DHEA uses the $\Delta 5$ pathway since there are very low levels of 3β-HSD in the zona reticularis. The presence of cytochrome b5 stimulates the 17,20 lyase reaction.
19. Adrenarche occurs around the age of 6 after the development of the zona reticularis. Peak adrenal androgen production is around 20-25 years of age.
20. Impairment of cortisol production by 21α-hydroxylase or 11β-hydroxylase deficiency leads to loss of feedback on the corticotropes in the anterior pituitary and consequently excess ACTH production. This expands the zona reticularis and leads to the over-production of adrenal androgens and atypical genitalia in newborns.
21. 11β-hydroxylase deficiency leads to salt retention due to the production of 11-DOC, which has mineralocorticoid activity. In contrast, 21α-hydroxylase deficiency leads to salt-wasting, as steroid precursors with mineralocorticoid activity are not manufactured.
22. 17α-hydroxylase deficiency leads to lack of cortisol and adrenal androgen production as well as hypertension from increased mineralocorticoid production.
23. The adrenal medulla synthesizes the catecholamines epinephrine and norepinephrine in a 4:1 ratio, as well as trace amounts of dopamine.
24. Virtually all circulating epinephrine originates from the adrenal medulla. Norepinephrine is also produced in post-ganglionic sympathetic nerve terminals around the body. Catecholamines are carried freely dissolved in the plasma.
25. Epinephrine is produced by the enzymatic processing of tyrosine in the adrenal medulla. Cortisol increases the expression of PNMT, which is the final step in epinephrine production.
26. The blood supply of the adrenal gland facilitates the action of cortisol on epinephrine synthesis, as blood flows from the cortex into the medulla, picking up cortisol as it flows through the zona fasciculata.
27. Cortisol stimulation of epinephrine synthesis enables a coordinated response to stress.